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ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Purification and Characteristics of Organic Compounds

(English Medium)

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PURIFICATION & CHARACTERISATION OF ORGANIC COMPOUNDS

METHODS OF PURIFICATION OF ORGANIC COMPOUNDS :

Organic compounds obtained either from natural source (or) synthesized in laboratory are contaminated with impuritics. Various methods are used for removal of impurities from an organic compound depends on the nature of compound and type of impurities present in it. The following methods are commonly used for purification

(1) Sublimation
(2) Crystallisation
(3) Distillation
(4) Solvent extraction (differential extraction)
(5) Chromatography.

Note: Most of the pure compounds contain sharp Melting point & Boiling points.

Sublimation :

The process of conversion of sublimable solid to vapour state directly by heating without passing through liquid state is called sublimation.

Solid vapour

- This method is used for purification of solids
- Sublimation process is used for separation of sublimable volatile compounds from non sublimable impurities.
- Sublimation is generally used for purification of camphor, napthalene. Anthracene. Benzoic acid, phthalic anhydride, Anthraquinone, Indigo and Iodine HgCl, solid SO_3

Crystallisation :

- It is used for purification of solid organic compounds.
- Crystallisation is based on the difference in solubilities of the compound and impurities in a suitable solvent.
- The principle involved in this method is impure compound dissolved in a solvent is sparingly soluble at low temperature, but appreciably soluble at high temperature. Insoluble impurities are separated by filtration. The filtrate on cooling saturated solution, pure compound crystallises out. If a compound is highly soluble in one solvent and too little soluble in another solvent then crystallisation is carried out by using mixture of these solvents.
- Impurities, which impart colour are removed by adsorbing over activated charcoal.
- Repeated crystallisation is required if organic compound contains impurities of comparable solubilities.
- The process of separation of different components of a mixture by repeated crystallisation is called fractional crystallisation.
- Fractional crystallisation is used for separation of two or more soluble substances which have different solubilities in the same solvent.
- Most commonly used solvents for crystallisation are water, alcohol, ether, chloroform, carbontetrachloride. acetone, benzene, petroleum ether.

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- Sugar having an impurity of common salt can be crystallised from hot ethanol. since sugar dissolves in hot ethanol but common salt does not.
- Fractional crystallisation can be used to separate a mixture of $KClO_3$ (less soluble) and KCl (more soluble).

DISTILLATION :

Distillation is an important method used to separate

i) Volatile liquids from non volatile impurities.

ii) Liquids having sufficient difference in boiling points.

Simple Distillation :

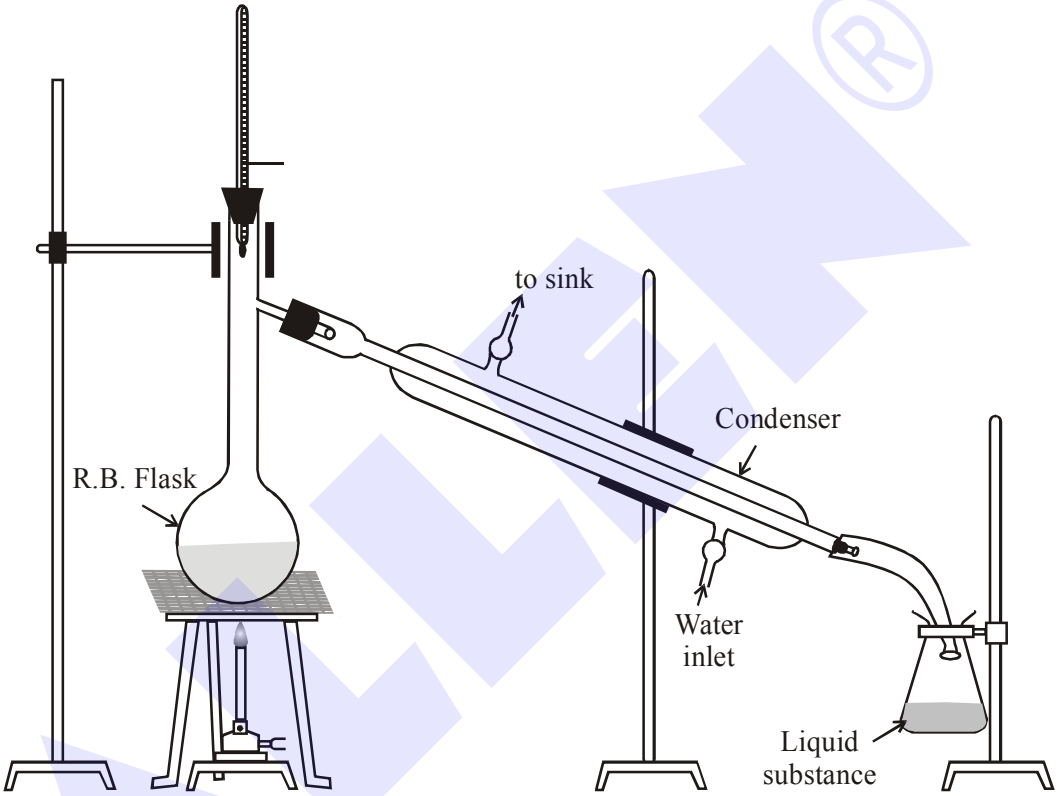


Fig : Simple distillation. The vapours of a substance formed are condensed and the liquid is collected in conical flask

- This process is used for purification of liquids which does not undergo decomposition at their boiling points
- The vapourisation of a liquid by heating and subsequent condensation of vapours by cooling is known as distillation.
- Liquid mixture is taken in a round bottom flask and heated carefully, the vapour component with lower boiling point distills first, the vapour formed is condensed by using condenser and the liquid is collected in a receiver. The vapours of component with higher boiling point distills later.
- The liquids that have boiling point difference greater than 25°C can be purified by this method,

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e.g:

(i) Chloroform (BP. 334 K) & Aniline (B.P. 457 K)

(ii) Ether (B.P. 308 K)& Toluene (B.P. 384 K)

(iii) Benzene (B.P. 353 K)& Aniline(B.P. 457 K)

Fractional Distillation :

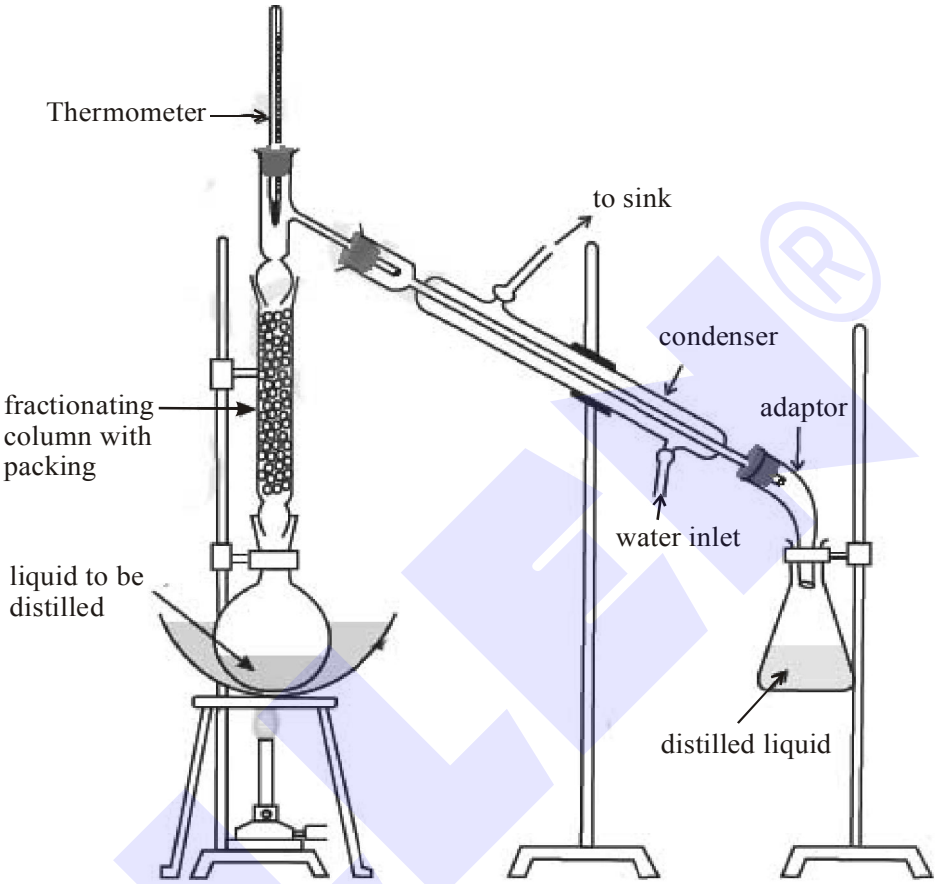


Fig. Fractional distillation : The vapours of lower boilling fraction reach the top of the column first followed by vapours of higher boilling fractions.

- Fractional distillation is used if the difference in boiling point of two liquids is less than 25° C.
- Vapours of liquid mixture are passed through fractionating column before condensation,which is fitted over mouth of the round bottom flask.
- Vapours of liquid with higher boiling point condense before the vapours of liquid with lower boiling point the vapours raising up in the fractionating column is richer in more volatile component.
- Fractionating column provides many surfaces for heat exchange between ascending vapours and descending condensed liquid.
- Each successive condensation and vapourisation unit in the fractionating column is called a theoretical plate.
- Liquids forming a constant boiling mixture (azeotropic mixture) can not be separated by this method.
- Fractional distillation is used to separate different fractions of crude oil in petroleum industry.
- This method is used for separation of mixture of acetone(B.P. 330K) and methyl alcohol (B.P. 338K)
- Mixture of benzene and toluene can be separated by fractional distillation.

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Ex.1 How is ethyl alcohol purified from methylated spirit.

Sol: Methylated spirit is ethyl alcohol contaminated mainly with methyl alcohol. Ethyl alcohol is purified by fractional distillation since the difference in boiling point is less.

DISTILLATION UNDER REDUCED PRESSURE (Vacuum Distillation)

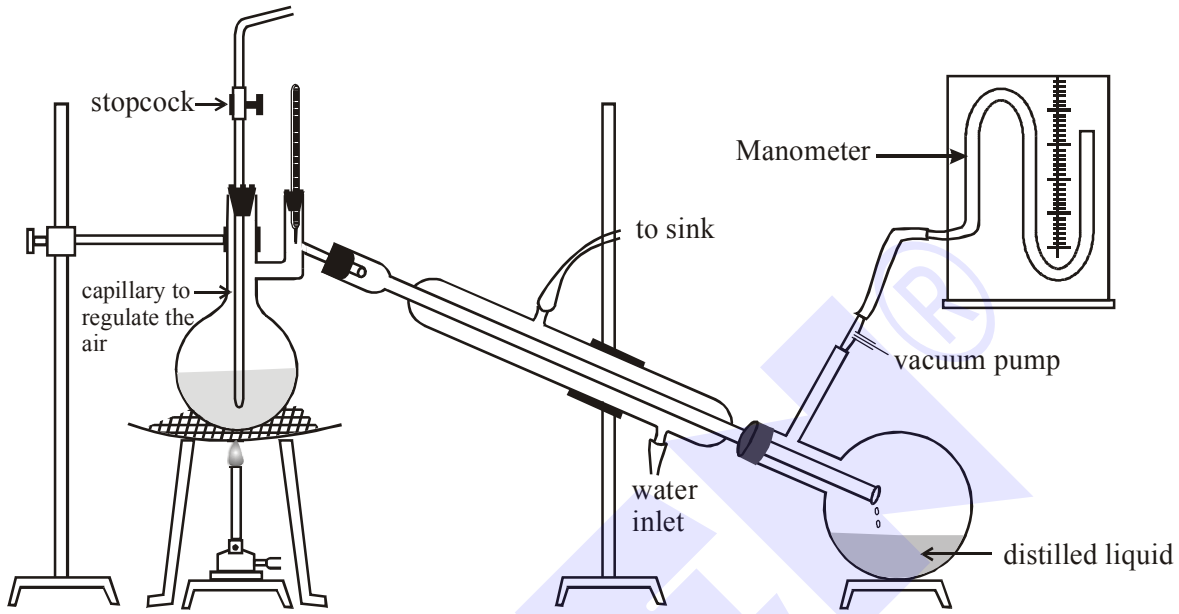


Fig. Distillation under reduced pressure : A liquid boils at a temperature below its vapour pressure by reducing the pressure

- This method is used to purify liquids having very high boiling points, which decompose at or below their boiling points.
- These liquids are made to boil at a temperature lower than their normal boiling point by reducing pressure on their surface with the help of vacuum pump.
- Glycerine, H_2O_2, formaldehyde are purified by vacuum distillation.
- Glycerol can be separated from spent-lye in soap industry by using vacuum distillation.
- Sugar cane juice is concentrated in sugar industry by evaporation under reduced pressure which saves lot of fuel.

Steam Distillation :

- This method is used for separation and purification of organic compounds (solids or liquids) which
  i) are steam volatile
  ii) are insoluble in water.
  iii) Process high vapour pressure (l0-15 mm of Hg at 373 K)
  iv) Contains non volatile impurities.
- Steam distillation is based on Dalton's law of partial pressure i.e., $P = P_1 + P_2$. where P = Atmospheric pressure.
  P_1 = Portial pressure of organic liquid
  P_2 = Vapour pressure due to water.

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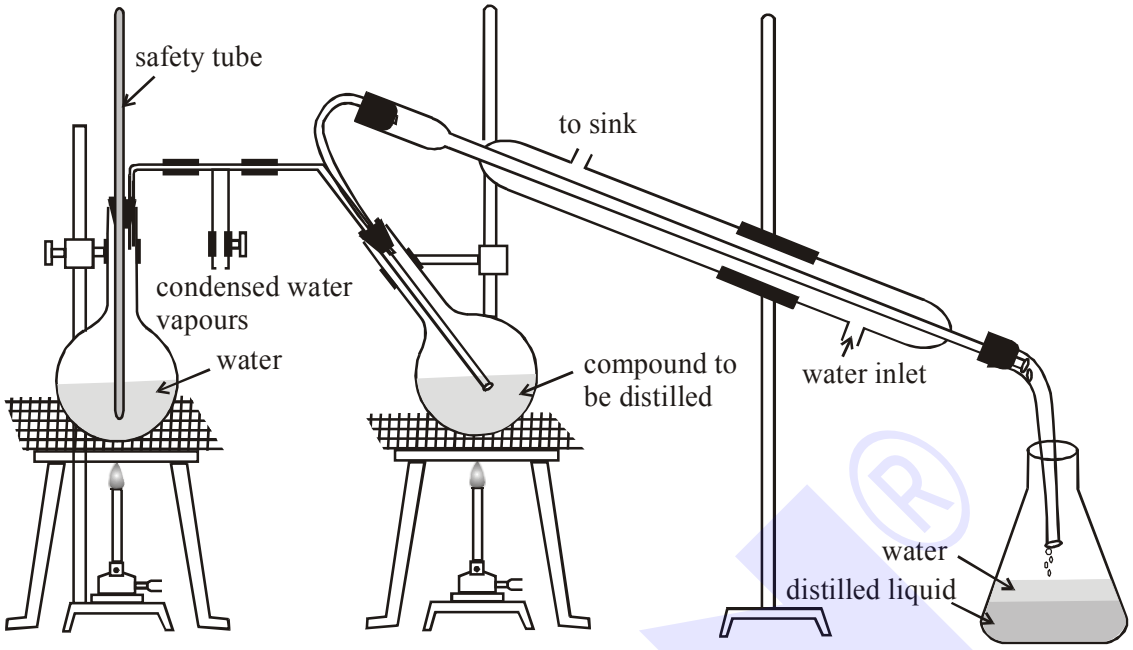


Fig. Steam distillation. Steam volatile component volatilizes, the vapours condense in the condenser and the liquid collects in conical flask

- Compounds which can be purified by steam distillation are aniline, nitrobenzene, bromobenzene, o-nitrophenol, o-hydroxy benzaldehyde (salicylaldehyde), o-hydroxy acetophenone, turpentine oil, essential oils.

Ex.2 Mention about the purification of (a) aniline and (b) naphthalene

Sol. (a) Aniline can be purified by steam distillation because it is immiscible with water and steam volatile.
(b) Naphthalene can be purified by sublimation because it changes on heating directly to vapour state and on cooling, it changes back into solid form.

Solvent Extraction (Differential Extraction) :

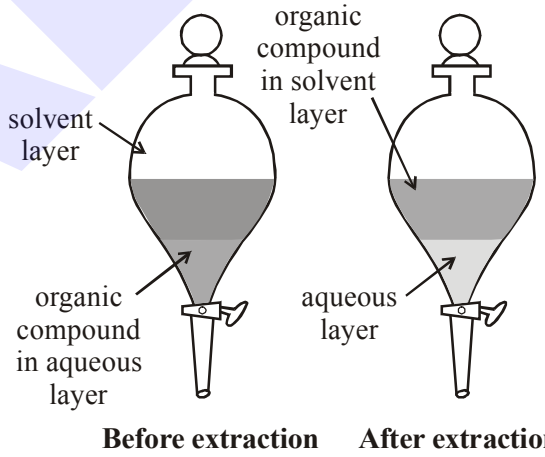


Fig. Differential extraction. Extraction of compound takes place based on difference in solubility

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- **Definition:** The process of isolating an organic compound from its aqueous solution by shaking with a suitable solvent is called differential extraction. It is also called solvent extraction
- When an organic compound is present in an aqueous medium, then it is separated by shaking it with an organic solvent in which it is more soluble than in water.
- Solvent should be immiscible with water and organic compound to be separated should be highly soluble in it
- Organic solvent and aqueous solution are immiscible with each other, so they can form two distinct layers (which can be separated by separator funnel).
- Organic solvent is distilled or evaporated to get organic compound.
- If organic compound is less soluble in organic solvent then large quantity of solvent is required to extract small quantity of compound, which is said to be continous extraction.
- Benzoic acid can be extracted from its aqueous solution using benzene as solvent.
- Ether is a better solvent in differential extraction due to:

  i) its less polarity

  ii) less reactivity

  iii) Less Boiling Point

  iv) higher solubility of organic compounds

SEPARATION BY CHEMICAL METHODS :

- It is used for mixture of substances which are chemically different.

e.g :

(1) Separation of acidic and basic compounds of coal-tar.

(2) $\underset{\text{(impure)}}{HC\equiv CH} \xrightarrow[Cu_2Cl_2]{\text{ammonical}} \underset{\text{(red ppt.)}}{CuC\equiv CCu} \xrightarrow[HCl]{\text{dil.}} \underset{\text{(pure)}}{HC\equiv CH}$

(3) $\underset{\text{Acid}}{\text{Pyroligneous}} \xrightarrow{Ca(OH)_2} \underset{\text{acetate}}{\text{Calcium}} \xrightarrow[HCl]{\text{conc.}} CH_3COOH$

(from wood distillation industry) ® $CH_3COOH + MeCoMe + MeOH$

(4) $\underset{\text{(Impure)}}{CH_3OH} \xrightarrow[\text{acid}]{\text{oxalic}} \underset{\text{(Crystalline)}}{\text{Methyl oxalate}} \xrightarrow[\Delta]{NaOH(aq)} \underset{\text{(pure)}}{CH_3OH}$

Chromatography :

- This method is used for separation of mixtures into their components, purification of compounds and also to test the purity of compounds.
- Chromatography is obtaind from the greek word "Chroma" means colour and "graphy" means writing.

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- This method was first used for separation of coloured substances found in plants.
- This method was described by Tswett.
- This Technique consists of two phases one is stationary phase of large surface area while the second is moving phase which is allowed to move slowly over the stationary phase.
- Stationary phase is either liquid or solid, while moving phase may be liquid or gas.
- The technique of chromatography is based on the rates at which the components of the mixture moves through a porous medium (called stationary phase) under the influence of some solvent (or) gas (called mobile phase).
- Mixture of substances is applied on a stationary phase which may be solid or liquid A pure solvent, a mixture of solvents or a gas is allowed to move slowly over the stationary phase, the components of the mixture get gradually separated from one another.
- Recovery of separated substances by using suitable solvent is known as elution. The solvent used is known as eluant.

S.No.	Chromatography process	Stationary Phase	Mobile Phase
1.	Column chromatography (Adsorption)	Solid	Liquid
2.	Thin layer chromatography (TLC)	Liquid (or) solid	Liquid
3.	Paper chromatography	Liquid	Liquid

- Based on the principle involved chromatography is classified in to

  a) Adsorption chromatography ex. Column chromatography and TLC

  b) partition chromatography ex. Paper chromatography

(a) **Adsorption Chromatography :**

- Adsorption chromatography is based on the fact that different compounds are adsorbed on an adsorbent to different degrees.
- Commonly used adsorbents are silica gel, alumina, magnesium oxide, cellulose powder, activated animal charcoal.
- When a mobile phase is allowed to move over stationary phase, the components of the mixture move by varying distances over stationary phase.
- There are two main types of chromatographic techniques based on principle of differential adsorption

  (i) Column chromatography and

  (ii) Thin layer chromatography (TLC)

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(i) Column Chromatography

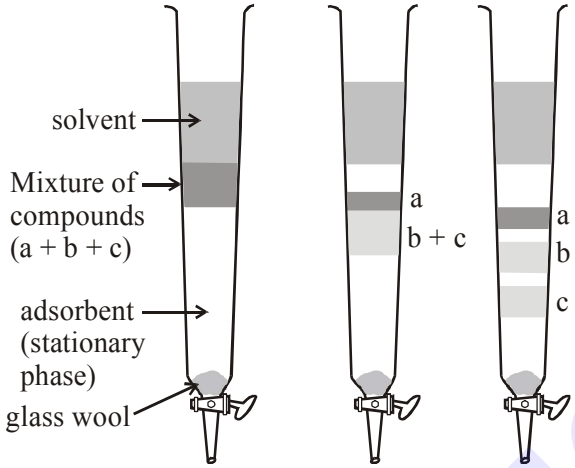


Fig. Column chromatography. Different stages of separation of components of a mixture.

- It involves separation of a mixture over a column of adsorbent packed in a glass tube, which is fitted with stop cock at its lower end.
- The mixture to be separated on the adsorbent is placed at the top of the stationary phase.
- An appropriate eluant, which is a liquid or a mixture of liquids is allowed to flow down the column slowly.
- The strongly adsorbed substances are retained near the top and others come down to various distances in the column,

(ii) Thin layer chromatography (TLC)

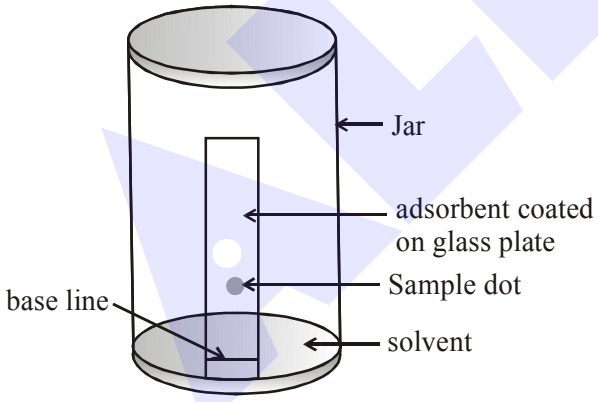


Fig. (a) thin layer chromatography.

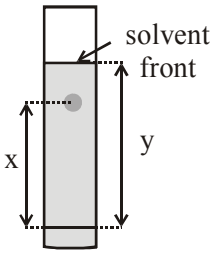


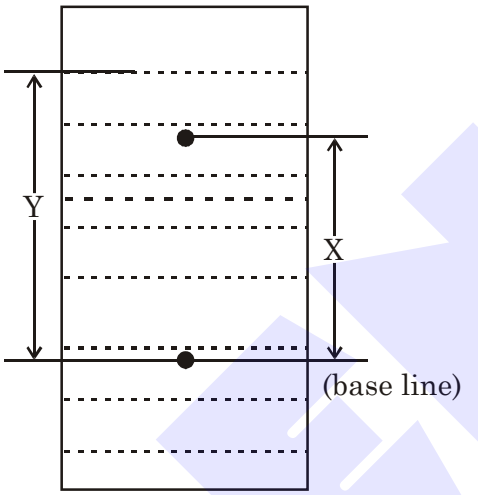
Fig. (b) Developed chromatograph

chromatogram being developed

- It involves separation of substances of a mixture over a thin layer of an adsorbent coated on glass plate.
- The glass plate is coated with adsorbent (ex: silica gel, alumina) as a thin layer (about 0.2mm thick) is called chromatography plate or chroma plate .

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- The solution of mixture to be separated is applied as small spot about 2cm above from one end of the TLC plate.
- The glass plate is placed in a closed jar containing the eluant. As the eluant rises up, the components of the mixture move up along the eluant to different distances depending on their degree of adsorption and separation takes place.
- The relative adsorption of each component of the mixture is expressed in terms of its retardation factor i.e, R_f value.



$$R_f = \frac{\text{Distance moved by the substance from base line (x)}}{\text{Distance moved by the solvent from base line (y)}}$$

- The spots of coloured compounds are visible on TLC plate due to their original colour.
- The colourless compound which fluroscene are detected with ultraviolet light
- Spots of compounds are even detected by allowing them to adsorb iodine, will show up as brown spots.
- Some times an appropriate reagent is sprayed on the plate.

eg : Amino acids are detected by spraying the plate with ninhydrin solution.

Partition Chromatography :

- Partition chromatography is based on continuous differential partitioning of components of a mixture between stationary and mobile phases.
- Paper chromatography is a type of partition chromatography.
- In paper chromatography a special quality known as chromatography paper is used.
- In Chromatography paper, cellulose helps as support, and water absorbed from air on to hydroxyl groups of cellulose acts as stationary phase.

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- The chromatography paper spotted with the solution of mixture at the base is suspended in a suitable solvent or mixture of solvents, this solvent (s) act as mobile phase.
- The solvent rises up the paper by capillary action and flows over the spot.
- The paper selectively retains different component according to their differing partition in the two phases. The paper strip so developed is called chromatogram.
- The spots of the separated coloured compounds are visible at different heights from the position of initial spot on the chromatogram.
- The spots of the separated colourless compound may be observed either under ultraviolet or by the use of appropriate spraying agent

Additional Information : Applications of Chromatography:

I. (i) In Chemical Industry column chromatography is used for separation of required components obtained after synthesis.

(ii)TLC is useful for monitoring large scale column chromatography.

II **Pharmaceutical industry :** Chromatography is used for separation of chiral compounds to obtained pharmaceutically active optical isomer.

III **Food Industry :** Chromatography techniques are used for quality control in food industry. It is used to determine presence and to separate additives, flavours etc. It is also used to detect presence of contaminents like mould, bacteria in food.

IV **Environment-Testing lab :** Presence and quality of pollutants in air and drinking water can be determined by chromatography technique.

V **Diagnostic Technique:** Presence of certain drugs and the marker compounds for medical diagnosis in blood and urine are determined.

Qualitative Analysis of Organic Compounds (Detection of Elements):

- The qualitative analysis of an organic compound involves detection of all elements present in it

Detection of Carbon and Hydrogen :

- Carbon and hydrogen are detected by heating the compound with cupric oxide (CuO).
- Carbon present in the compound is oxidised to carbondioxide, which turns lime water milky.
- Hydrogen present in the compound is converted into water, which turns anhydrous copper sulphate into blue.

$$C + 2CuO \xrightarrow[\Delta]{} 2Cu + CO_2$$

$$H_2 + CuO \xrightarrow[\Delta]{} Cu + H_2O$$

- Carbondioxide turns lime water milky.

$$Ca(OH)_2 + CO_2 \rightarrow \underset{\text{(milky)}}{CaCO_3} + H_2O$$

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- Water vapours turn anhydrous copper sulphate into blue.

$$\underset{\text{anhydrous (colourless)}}{CuSO_4} + 5H_2O \longrightarrow \underset{\text{hydrated (blue)}}{CuSO_4.5H_2O}$$

Detection of Nitrogen, Sulphur Halogens & Phosphorus :

- Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by Lassaigne's test.
- Organic compounds are fused with dry sodium in fusion tube and fused mass after extraction with water is boiled and filtered, the filtrate is called sodium fusion extract
- During prepartion of sodium fusion extract covalent compound is converted into ionic compound.
- The following reactions takes place

$$Na + C + N \xrightarrow{\Delta} NaCN$$

$$2Na + S \xrightarrow{\Delta} Na_2S$$

$$2Na + X_2 \xrightarrow{\Delta} 2NaX \ (X = Cl, Br \text{ or } I)$$

Test for Nitrogen :

Sodium fusion extract is boiled with freshly prepared ferrous sulphate ($FeSO_4$) solution and then acidified with concentrated sulphuric acid. The formation of Prussian blue colour confirms the presence of nitrogen.

$$2NaCN + FeSO_4 \rightarrow Fe(CN)_2 + Na_2SO_4$$

$$Fe(CN)_2 + 4NaCN \rightarrow Na_4[Fe(CN)_6]$$

sodium hexacyanoferrate(II)

On heating with concentrated sulphuric acid some Iron(II) ions are oxidised to Iron(III) ion, which reacts with sodium hexacyanoferrate(II) to produce Iron (III) hexacyanoferrate(II) (ferri ferrocyanide) which is Prussian blue in colour.

$$3Na_4[Fe(CN)_6] + 2Fe_2(SO_4)_3$$

$$\rightarrow Fe_4[Fe(CN)_6]_3 + 6Na_2SO_4$$

ferriferrocyanide

(prussianblue)

- This test fails in case of diazo compounds.
- If the amount of nitrogen present is less, then Prussian blue is present in collaidal form and the solution looks green.

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Ex.3 Hydrazine does not give Lassaigne's test, Why?

Sol. In the Lassaigne's test, nitrogen is converted to cyanide by combining with carbon of the compound Hydrazine does not contain carbon and hence cyanide cannot form.

Ex.4 Why diazonium salts do not show positive Lassaigne's test for nitrogen ?

Sol: Diazonium salts are unstable and lose nitrogen as N_2 gas on heating. Hence during fusion, no sodium cyanide is formed in Lassaigne's extract due to the loss of nitrogen.

Test for Sulphur:

(a) Sodium fusion extract is acidified with acetic acid and lead acetate is added to it, a black precipitate of lead sulphide is formed, which indicates presence of sulphur.

$$Na_2S + (CH_3COO)_2Pb \rightarrow \underset{\text{black}}{PbS\downarrow} + 2CH_3COONa$$

(b) Sodium fusion extract is treated with freshly prepared sodium nitroprusside, appearance of violet colour (purple) indicates presence of sulphur.

$$Na_2S + Na_2[Fe(CN)_5NO] \rightarrow \underset{\text{(violet)}}{Na_4[Fe(CN)_5NOS]}$$

(sodium nitro prusside)

- In case both nitrogen and sulphur arc present in an organic compound sodium thiocyanate is formed, which gives blood red colour with neutral $FeCl_3$ solution.

$$Na + C + N + S \rightarrow NaSCN$$

$$3NaSCN + FeCl_3 \rightarrow \underset{\text{(blood red)}}{Fe(SCN)_3} + 3NaCl$$

(or)

$$Fe^{+3} + SCN^- \rightarrow \underset{\text{(blood red)}}{[Fe(SCN)]^{+2}}$$

- If sodium fusion is carried out with excess of sodium, the thiocyanate decomposes to yield cyanide and sulphide, these ions gives their usual tests.

$$NaSCN + 2Na \rightarrow NaCN + Na_2S$$

Test for Halogens:

- Sodiumfusion extract is acidified with nitric acid and then treated with silver nitrate solution.

(i) White precipitate, soluble in ammonium hydroxide indicates presence of chlorine.

$$NaCl + AgNO_3 \rightarrow \underset{\text{(white ppt)}}{AgCl\downarrow} + NaNO_3$$

$$AgCl + 2NH_4OH \rightarrow \underset{\text{(soluble complex)}}{[Ag(NH_3)_2]Cl} + 2H_2O$$

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(ii) Yellowish precipitate, sparingly soluble in ammonium hydroxide indicates presence of bromine.

$NaBr + AgNO_3 \rightarrow AgBr \downarrow + NaNO_3$

(yellowish ppt)

(iii) Yellow precipitate, insoluble in ammonium hydroxide indicates presence of Iodine.

$NaI + AgNO_3 \rightarrow AgI \downarrow + NaNO_3$

yellow
precipitate

- Nitrogen and sulphur are also present in the compound, the sodum fusion extract is boiled with concentrated nitric acid to decompose sodium cyanide & sodium sulphide formed during Lassaigne's test, otherwise they interfere with silver nitrate test for halogens.

$NaCN + HNO_3 \rightarrow NaNO_3 + HCN \uparrow$

$Na_2S + 2HNO_3 \rightarrow 2NaNO_3 + H_2S \uparrow$

- If NaCN and Na_2S are not decomposed, then white and black precipitates of AgCN and Ag_2S are formed respectively with silver nitrate solution.

Beilstein's Test:

A copper wire flattened at one end is heated in an oxidising flame of Bunsen burner. The heating is continued till it does not impart blue colour flame. The hot end of copper wire is now touched with the organic substance and is once again kept in flame, the appearance of green or blue colour indicates the presence of halogens in the organic compound.

Limitations :

(a) Substances such as urea, thiourea do not contain halogens but gives this test

(b) It does not tell which halogen is present in organic compound.

Test for Phosphorus :

The compound is heated with oxidising agent (sodium peroxide) or with fusion mixture (sodium carbonate and potassium nitrate) phosphorus present in the compound is oxidised to sodium phosphate. The solution is boiled with nitric acid and then treated with ammonium molybdate, a canary yellow (ammonium phospho molybdate) precipitate formation (or) yellow colouration indicates presence of phosphorus.

$2P + 5Na_2O_2 \xrightarrow{\Delta} 2Na_3PO_4 + 2Na_2O$

$Na_3PO_4 + 3HNO_3 \longrightarrow H_3PO_4 + 3NaNO_3$

$H_3PO_4 + 12(NH_4)2MoO_4 + 21HNO_3 \longrightarrow$

$(NH_4)_3PO_4.12MoO_3 + 21NH_4NO_3 + 12H_2O$
(ammonium phospho molybdate)

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QUANTITATIVE ANALYSIS

It involves the estimation of percentage composition of various elements by suitable methods.

Estimation of Carbon and Hydrogen

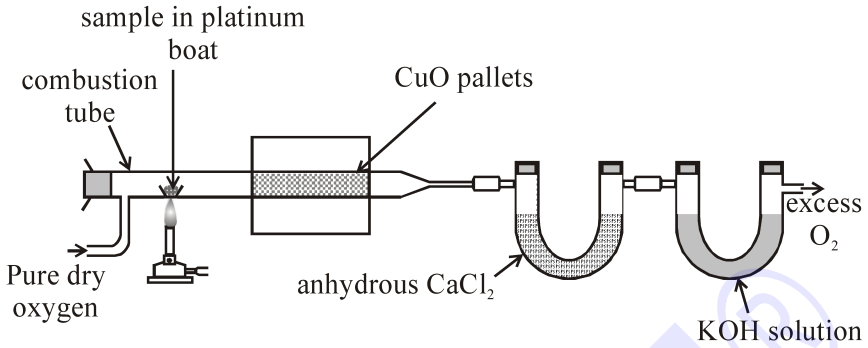


Fig. Estimation of carbon and hydrogen. Water carbon dioxide formed on oxidation of substance are absorbed in anhydrous calcium chloride and potassium hydroxide solutions respectively contained in U tubes

- Carbon and hydrogen are estimated by Liebig's combustion method.
- A known mass of an organic compound is burnt in the presence of excess of oxygen (free from CO_2) and Cupric oxide (CuO)
- Carbon and hydrogen present in the compound are oxidised to CO_2 and H_2O respectively

$$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \xrightarrow{\Delta} xCO_2 + \frac{y}{2}H_2O$$

- CO_2 and H_2O produced are weighed by absorbing in concentrated solution of potassium hydroxide and anhydrous calcium chloride (or) magnesium perchlorate respectively

$$\%C = \frac{12}{44} \times \frac{\text{weight of } CO_2 \text{ formed}}{\text{weight of organic compound}} \times 100$$

$$\%H = \frac{2}{18} \times \frac{\text{weight of } H_2O \text{ formed}}{\text{weight of organic compound}} \times 100$$

Ex.5 On complete combustion, 0.246g of an organic compound gave 0.198 g of carbon dioxide and 0.1014g of water. Determine the percentage composition of carbon and hydrogen in the compound.

Sol. $$\%\text{ of carbon} = \frac{12 \times 0.198 \times 100}{44 \times 0.246} = 21.95$$

$$\%\text{ of hydrogen} = \frac{2 \times 0.1014 \times 100}{18 \times 0.246} = 4.58$$

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ESTIMATION OF NITROGEN

Nitrogen present in organic compound is estimated by

(a) Dumas method (b) Kjeldahl's method

(a) Dumas method

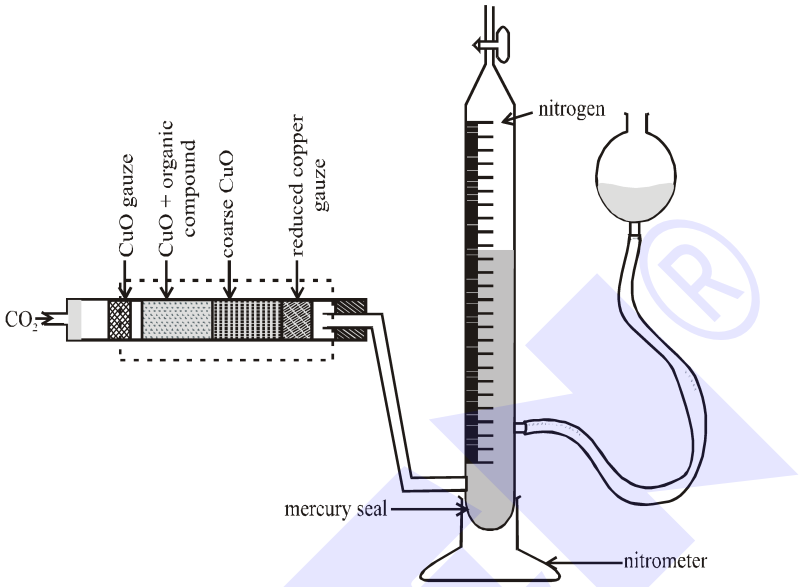


Fig. Dumas method. The organic compound yields nitrogen gas on heating it with copper(II) oxide in the presence of CO_2 gas. The mixture of gases is collected over potassium hydroxide solution in which CO_2 is absorbed and volume of nitrogen gas is determined.

- In this method nitrogen present in the organic compound is converted in to N_2(molecular nitrogen)
- A weighed amount of organic compound is heated with cupric oxide in an atmosphere of carbondioxide.
- Carbon and hydrogen present in the compound are oxidised to CO_2 and H_2O, while N_2 is at free.
- Some oxides of nitrogen formed are reduced to free nitrogen by passing over heated copper gauze

$$C_xH_yN_z + \left(2x + \frac{y}{2}\right)CuO \rightarrow$$

$$xCO_2 + \frac{y}{2}H_2O + \frac{z}{2}N_2 + \left(2x + \frac{y}{2}\right)Cu$$

- Oxides of nitrogen + Cu $\rightarrow$ N_2 + CuO
- The mixture of gases produced is collected over caustic potash solution (KOH solution) which absorbs CO_2.
- Nitrogen is collected in the upper part of nitro meter.

$$\%N = \frac{28}{22400} \times \frac{\text{Volume of nitrogen in ml at STP}}{\text{Weight of organic compound}} \times 100$$

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Ex.6. 0.25g of an organic compound gave 30 cm^3 of moist dinitrogen at 288 K and 745 mm pressure. Calculate the percentage of nitrogen. (Aqueous tension at 288 K = 12.7 mm)

Sol. Mass of the substance = 0.25 g

Volume of moist dinitrogen = 30 cm^3

Temperature = 288 K

Pressure = 745 – 12.7 = 732.3 mm

Volume of dinitrogen at STP :

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2} = \frac{732.3 \times 30 \times 273}{288 \times 760} = 27.4\,cm^3$$

$$\text{Percentage of nitrogen in organic compound} = \frac{28}{22400} \times \frac{\text{vol of } N_2 \text{ at STP}}{\text{wt of organic compound}} \times 100$$

$$\frac{28}{22400} \times \frac{27.4}{0.25} \times 100 = 13.6$$

Ex.7 0.3g of an organic compound gave 50 ml nitrogen at 27°C and 715 mm pressure. If the aqueous tension at 27°C is 15 mm, calculate the percentage composition of nitrogen in the compound.

Sol. Mass of the substance = 0.3 g

Volume of the moist dinitrogen = 50 ml

Temperature = 27°C = 27 + 273 = 300 K

Pressure = 715 – 15 = 700 mm

Volume of dintrogen at STP :

$$v_2 = \frac{P_1V_1}{T_1} = \frac{T_2}{P_2} = \frac{700 \times 50 \times 273}{760 \times 300} = 41.9 \text{ ml}$$

$$\frac{28}{22400} \times \frac{\text{vol of } N_2 \text{ STP}}{\text{wt of organic compound}} \times 100 = \frac{28}{22400} \times \frac{41.9}{0.3} \times 100 = 17.46$$

22400 ml of dinitrogen at STP weight = 28 g

41.9 ml dinitrogen at STP weight = $\frac{28 \times 41.9}{22400}$ g

0.3 g of organic compound contains $\frac{28 \times 41.9}{22400}$ g of N_2

100 g of organic compound contain $\frac{28 \times 41.9}{22400} \times \frac{100}{0.3} = 17.46$

percentage of nitrogen = 17.46

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Kjeldahl's Method :

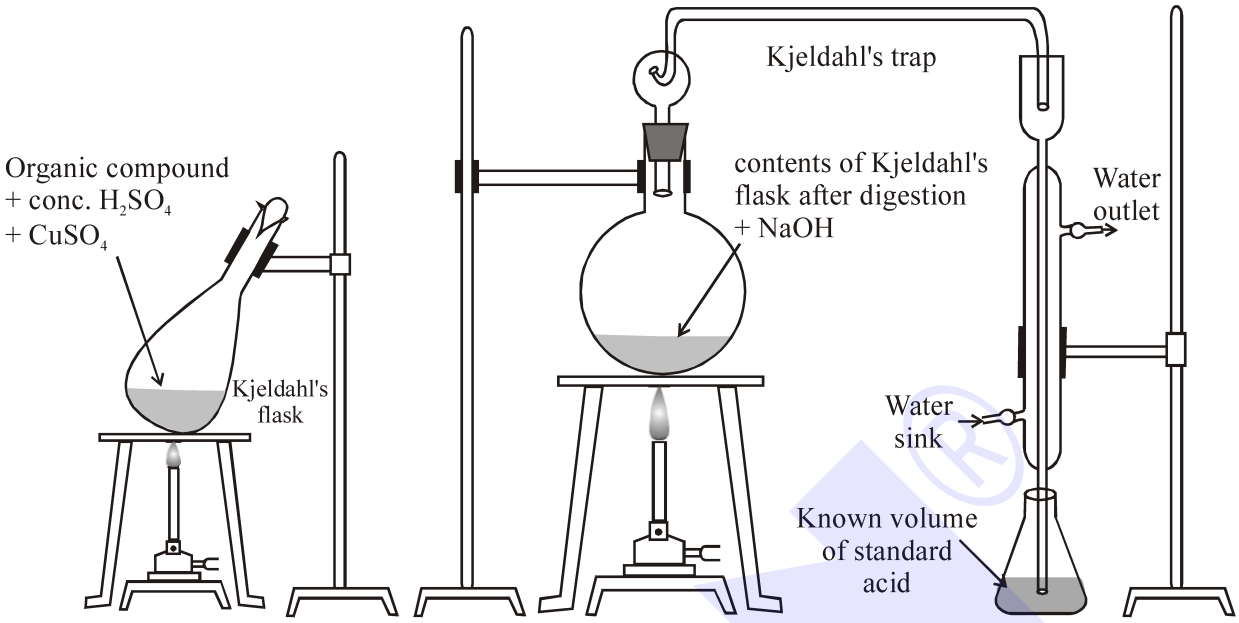


Fig. Kjeldahl method. Nitrogen-containing compound is treated with concentrated H_2SO_4 to get ammonium sulphate which liberates ammonia on treating with NaOH; ammonia is absorbed in known volume of standard acid.

- In this method nitrogen in the organic compound is converted into ammonia (NH_3)
- A known mass of organic compound containing nitrogen is heated with concentrated sulphuric acid in presence of K_2SO_4 and $CuSO_4$ then nitrogen present in the compound is converted into ammonium sulphate. K_2SO_4 increases boiling point of H_2SO_4 and $CuSO_4$ acts as catalyst.
- Organic compound + $H_2SO_4 \longrightarrow (NH_4)_2SO_4$
- The resulting solution is distilled with excess of sodium hydroxide

  $(NH_4)_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2NH_3 + 2H_2O$
- Ammonia evolved is absorbed in a known but excess volume of standard HCl (or) H_2SO_4 solution.
- The acid left unreacted is estimated by titration against standard solution of sodium hydroxide.

$$\text{Percentage of Nitrogen} = \frac{14}{1000} \times \frac{V \times N}{\text{wt. of organic compound}} \times 100$$

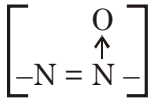
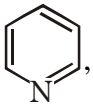
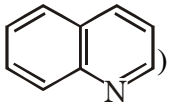
$$\%N = \frac{1.4 \times V \times N}{\text{Weight of organic compound}}$$

Where V = volume of acid in ml neutralised by ammonia

N = Normality of acid.

- This method is simpler and more convenient
- It is mainly used to find percentage of nitrogen present in food stuffs, soils, fertilizers and various agricultural products.

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- This method is not applicable to compounds containing nitro ($-NO_2$), Nitroso (NO), azo group ($-N=N-$), azoxy compounds  and nitrogen present in the ring (pyridine  , quinoline ) because nitrogen present in these compounds is not quantitatively converted in to ammonium sulphate.

Ex.8 In Kjeldahl's estimation of nitrogen, the ammonia evolved from 0.5 g of an organic compound neutralised 10 ml of 1M H_2SO_4. Calculate the percentage of nitrogen in the compound?

Sol. 10 ml of 1M H_2SO_4 = 20ml of 1M NH_3

1000 ml of 1M ammonia contains 14 g nitrogen

20 ml of 1M ammonia contains $\frac{14\times20}{1000}$ g N_2

$$\% \text{ of nitrogen} = \frac{14\times20\times100}{1000\times0.5} = 56.0$$

Ex.9 In Kjeldahl's estimation of nitrogen, the ammonia obtained from 0.5 g of an organic substance was passed into 100 cm^3 of $\frac{M}{10}$ NaOH for neutralisation. Calculate the percentage of nitrogen in the compound.

Sol. m eq. of H_2SO_4 taken = molarity x basicity x volume (ml)

$$= \frac{1}{10}\times2\times100 = 20$$

$$\text{m.eq of NaOH} = \frac{1}{10}\times1\times154 = 15.4$$

m. eq of H_2SO_4 unused - 20 – 15.4 = 4.6

$$\% \text{ of nitrogen} = \frac{1.4\times\text{meq of } H_2SO_4 \text{ unused}}{\text{wt of organic compound}} = \frac{1.4\times4.6}{0.5} = 12.88\%$$

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ESTIMATION OF HALOGENS CARIUS METHOD

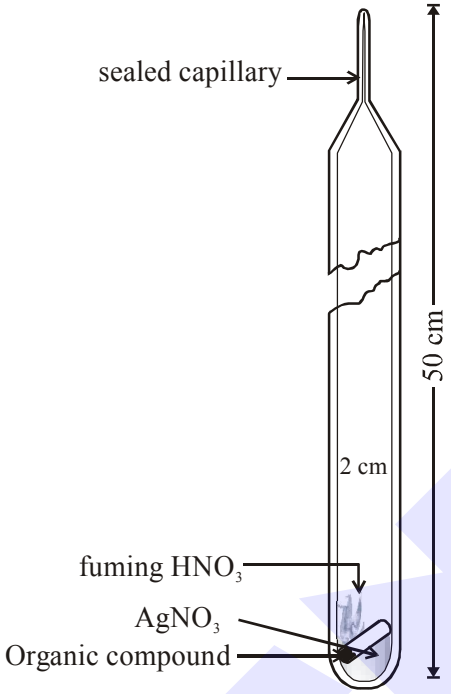


Fig. Carius method. Halogen containing organic compounds is heated with fuming nitric acid in the presence of silver nitrate

- A weighed amount of an organic compound is heated with fuming nitric acid in the presence of silver nitrate contained in a hard glass tube known as carius tube.
- Carbon and hydrogen present in the compound is converted into CO_2 and H_2O.
- Halogen present in the organic compound is converted into silver halide.
- The perecipitate is washed, dried and weighed

$$\text{Percentage of halogen} = \frac{\text{Atomic weight of halogen}}{\text{Mwt of silver halide}} \times \frac{\text{Weight of silver halide formed}}{\text{Weight of organic compound}} \times 100$$

$$\%\text{Cl} = \frac{35.5}{143.5} \times \frac{\text{wt of AgCl formed}}{\text{wt of organic compound}} \times 100$$

$$\%\text{Br} = \frac{80}{188} \times \frac{\text{wt. of AgBr formed}}{\text{Wt. of organic compound}} \times 100$$

$$\%\text{I} = \frac{127}{235} \times \frac{\text{Wt. of AgI formed}}{\text{Wt. of organic compound}} \times 100$$

Ex.10 In Carius method, 0.1890g of an organic compound gave 0.2870g of silver chloride. Calculate the percentage of chlorine in the compound

Sol: Weight of substance = 0.1890g

Weight of silver chloride = 0.2870g

$$\%\text{ of chlorine} = \frac{\text{weight of AgCl} \times 35.5 \times 100}{\text{weight of substance} \times 143.5} = \frac{0.2870 \times 35.5 \times 100}{0.1890 \times 143.5} = 37.8$$

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Ex.11 One gram of bromoalkane on heating with excess silver nitrate in Carius tube method gave 0.94g of yellow precipitate. What is the percent weight of halogen ?

Sol. Weight of substance = lg Yellow precipitate is AgBr ; Weight of AgBr = 0.94g

$$\% \text{ of bromine} = \frac{0.94 \times 80 \times 100}{1 \times 188} = 40$$

Ex.12 In carius method of estimation of halogen 0.15g of an organic compound gave 0.12g of AgBr. Find out the percentage of bromine in the compound

Sol. $$\% \text{ of bromine} = \frac{80 \times 0.12 \times 100}{188 \times 0.15} = 34.04$$

Estimation of Sulphur Carius method:

- A weighed amount of organic compound is heated in a carius tube with sodium peroxide or fuming nitric acid.
- Sulphur present in the compound is oxidised into sulphuric acid, which is treated with $BaCl_2$ solution gives precipitate of $BaSO_4$.
- It is filtered, the precipitate is washed, dried and weighed.

$$\%S = \frac{\text{Atomic wt of sulphur}}{\text{Mol. wt of } BaSO_4} \times \frac{\text{wt of } BaSO_4 \text{ formed}}{\text{wt of organic compound}} \times 100$$

$$\%S = \frac{32}{233} \times \frac{\text{wt of } BaSO_4 \text{ formed}}{\text{wt. of organic compound}} \times 100$$

Ex.13 In sulphur estimation, 0.157 g of an organic compound gave 0.4813 g of barium sulphate. What is the percentage of sulphur in the compound?

Sol. 233 g $BaSO_4$ contain 32g sulphur

0.4813g $BaSO_4$ contain $\frac{32 \times 0.4813}{233}$ g sulphur

$$\% \text{ of sulphur} = \frac{32 \times 0.4813 \times 100}{233 \times 0.157} = 42.10$$

Ex.14 On heating 0.2g of an organic compound with a mixture of barium chloride and nitric acid, 0.466g of barium sulphate was obtained. Calculate the percentage of sulphur.

Sol. Weight of substance = 0.2 g

Weight of barium sulphate = 0.466 g

$$\text{Weight percentage of sulphur} = \frac{0.466 \times 32 \times 100}{0.2 \times 233} = 32$$

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Estimation of Phosphorus Carius Method :

A weighed amount of organic compound is heated with fuming nitric acid, then phosphorus present in the compound is oxidised to phosphoric acid. Phosphoric acid is precipitated as magnesium ammonium phosphate ($Mg\,NH_4\,PO_4$), by addition of magnesia mixture ($MgCl_2 + NH_4OH + NH_4Cl$)

- Magnesium ammonium phosphate is washed, dried and it is heated strongly to get magnesium pyrophosphate {$Mg_2P_2O_7$).

$$\underset{\text{(Phosphoric acid)}}{H_3PO_4} \xrightarrow{\text{Magnesia mixture}} \underset{\text{(Magnisium ammonium phosphate)}}{MgNH_4PO_4}$$

$$2MgNH_4PO_4 \xrightarrow{\Delta} \underset{\text{(Magnesium Pyrophosphate)}}{Mg_2P_2O_7} + 2NH_3 + H_2O$$

$$\%\,P = \frac{62}{222} \times \frac{\text{wt. of } Mg_2P_2O_7 \text{ formed}}{\text{wt. of organic compound}} \times 100$$

- Phosphoric acid is precipitated as ammonium phospho molybdate $(NH_4)_3PO_4.12MoO_3$ by adding ammonia and ammonium molybdate. (Molecular mass of $(NH_4)_3PO_4.12MoO_3 = 1877$)

$$\%P = \frac{31}{1877} \times \frac{\text{wt of } (NH_4)_3\,PO_4.12MoO_3 \text{ formed}}{\text{wt of organic compound}} \times 100$$

Estimation of oxygen .

- Usually percentage of oxygen in organic compound is determined by method of difference % of oxygen = 100 – (sum of the percentages of all other elements)
- Oxygen present in the organic compound is estimated by Aluise's method.
- A known amount of organic compound is subjected to pyrolysis in a stream of nitrogen.
- The mixture of gaseous products containg oxygen is passed over red-hot coke, then all the oxygen is converted in to carbon monoxide.

$$\text{compound} \xrightarrow{\Delta} O_2 + \text{other gaseous products}$$

$$2C + O_3 \xrightarrow{1373K} 2CO$$

- CO formed is quantitatively converted in to CO_2 by passing over warm Iodine pentoxide (I_2O_5)

$$5CO + I_2O_5 \longrightarrow I_2 + 5CO_2$$

The resulting gaseous mixture (CO_2 and I_2) is passed through potassium iodide solution, which absorbs iodine, and then passed over KOH to absorb CO_2.

$$\%\,\text{of oxygen} = \frac{16}{44} \times \frac{\text{wt. of } CO_2 \text{ formed}}{\text{wt. of organic compound}} \times 100$$

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Ex.15 0.2g of an organic compound on analysis gave 0.147g of carbondioxide, 0.12g of water and 74.6 c.c of nitrogen at S.T.P. Calculate the weight percentages of constituents.

Sol. Weight of compound = W= 0.2g

Weight of $CO_2 = W_1 = 0.147g$

Weight of $H_2O = W_2 = 0.12g$

Volume of N_2 at STP = 74.6 c.c.

$$\% \text{ of carbon} = \frac{W_1 \times 12 \times 100}{W \times 44} = \frac{0.147 \times 12 \times 100}{0.2 \times 44} = 20.04\%$$

$$\% \text{ of Hydrogen} = \frac{W_2 \times 2 \times 100}{W \times 18} = \frac{0.12 \times 2 \times 100}{0.2 \times 18} = 6.66\%$$

$$\% \text{ of Nitrogen} = \frac{W_2 \times 28 \times 100}{W \times 22400} = \frac{74.6}{8 \times 0.2} = 46.63\%$$

Remaining is oxygen.

% of Oxygen = 100 – (%C + %H + %N) = 100 – 73.33 = 26.67%

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